TRIBHUVAN UNIVERSITY

Institute of Science and Technology



A Dissertation Proposal Report Submission in

Partial fulfillment of the requirement for the Master's degree in Information Technology

Optimization of Unsupervised Image - to - Image translation using Semantic Based Techniques.

Under the Supervision of

……..

Submitted By:

Avishek Kuinkel
TU Roll number (7925002)

Submitted To:

Central Department of Computer Science and Information Technology

Tribhuvan University

Kirtipur, Kathmandu, Nepal

November, 2025

Table of Contents

List of Figure iii
List of abbreviation iv
CHAPTER 1: Introduction 1
1.1 Introduction 1
1.2. Problem Statement 2
1.3. Objective 2
CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW 3
2.1. Background Study 3
2.2. Literature Review 5
CHAPTER 3: METHODOLOGY 7
3.1 Description of Methodology 7
3.2 Flowcharts Depicting Methodology 8
3.3 Description of Data 9
3.4 Data Pre-processing 9
3.5 Description of Algorithms/ Mathematical Models 10
    3.5.1. DualGan 10
3.6. Performance Evaluation Metrics 13
Frechet Inception Distance (FID) 14
CHAPTER 4: WORKING SCHEDULE AND EXPECTED OUTPUT 15
4.1 Working Schedule 15
4.2 Expected Output 15
References 16

List of Figure

Figure 3. 1. Flowcharts of Depicting Methodology..... 8
Figure 3. 2.Network Architecture and data flow chart of Dual-GAN for I2I translation 10
Figure 4. 1.Gantt Chart..... 15

List of abbreviation

CGAN	conditional Generative Adversial Network
Dual-GAN	Dual Generative Adversial Network
I2I	Image-to-Image Translation
WGAN	Wasserstein Generative Adversarial Network

CHAPTER 1: Introduction

1.1 Introduction

Image-to-image translation is the task of taking images from one domain and transforming them so they have the style (or characteristics) of images from another domain. I2I has drawn increasing attention and made tremendous progress in recent years because of its wide range of applications in many computer vision and image processing problems, such as image synthesis, segmentation, style transfer, restoration, and pose estimation. I2I has become a key technology in computer vision, crucial for tasks ranging from style transformation to domain adaptation. I2I has drawn a lot of attention and made tremendous progress in recent years because of its wide range of applications in many computer vision [1]. If we take a selfie and want to make it more artistic as a drawing from a cartoonist, we can automatically achieve that with a computer through the use of image-to-image translation (I2I).

I2I is also used for converting images from low resolution to high-resolution, blurry-to -sharp, Thermal-to-RGB, Rainy-to-summer, sketch-to-photo etc. In general, the goal of I2I is to convert an input image from a source domain A to a target domain B with the intrinsic source content preserved and the extrinsic target style transferred. Generally, supervised methods usually produce better translated results than unsupervised methods on similar network structure and labeled data. Semantic segmentation is an important task in computer vision that aims to infer pixel-level semantic label information in images. Recently significant progresses have been made in deep neural networks-driven segmentation techniques, but these often require a large number of labels to supervise neural network training. It seems that there is a need for the unsupervised Semantic Based I2I translation to reduce lots of manual tasks in image labelling and to obtain the better results in Unsupervised I2I translation. cGAN achieved a huge success as a general-purpose solution to I2I translation problems. With human labeling, the expense is high, even impractical, and large quantities of data may not always be available.

Dual-GAN enables image translators to be trained from two sets of unlabeled images from two domains. Despite its potential, It still struggle with tasks involving complex semantic labels, particularly for tasks requiring semantic segmentation, which often relies on voluminous labeled datasets [1].

1.2. Problem Statement

Obtaining a sufficient number of labeled training data is challenging and sometimes impractical in real-world applications. Unsupervised image-to-image (I2I) interpretation faces challenges due to the reliance on expansive datasets of labeled pictures for preparing conditional GANs (cGANs). While the DualGAN approach enables training with unlabeled images from different domains, it struggles with tasks involving complex semantic labels compared to cGANs.

This research aims to overcome the limitations of DualGAN. The focus is on investigating the impact of a 'warm start' phase, which uses a limited set of labeled data, on Dual-GAN's ability to accurately maintain pixel and label correspondence in I2I translation. This strategy is inspired by the semi-supervised learning approaches involving real and pseudo labels and which extends GANs to semi-supervised learning by modifying the discriminator to also classify inputs into specific classes.

The research primarily seeks to boost DualGAN's capacity for semantic label translation by evaluating the effectiveness of integrating labeled data in the initial training phase and its subsequent influence on unsupervised translation. The goal is to enhance DualGAN's semantic translation performance, potentially advancing the field of unsupervised I2I translation.

1.3. Objective

1. To significantly reduce reliance on large datasets, making unsupervised I2I translation more practical in scenarios with scarce annotations.
2. To improve the performance of DualGAN for semantic-based image-to-image translation by leveraging labeled data as a pre-training step.

CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW

2.1. Background Study

The concept of image to image (I2I) interpretation has experienced noteworthy advancement, getting to be a foundation within the field of computer vision. This advancement reflects a move from conventional picture preparing methods, frequently including manual mediation, to more modern, mechanized strategies empowered by headways in machine learning and neural systems [2]. I2I translation's core objective is to convert a picture from one space to another while protecting its inborn substance, an errand that has seen significant refinement and development over a long time.

GAN, is a generative model that has become widely discussed within the machine learning community. It has significantly advanced a variety of computer vision tasks, as evidenced by its achievements in image processing, style transfer, classification, and image generation. The principle of GAN is inspired by the minimax two-person zero-sum game, where one player's gain is exactly equal to the other's loss. In the context of GANs, these 'players' are identified as the generator and the discriminator. The generator captures the information dispersion from genuine tests and maps it to a unused space, whereas the discriminator assesses whether the input, either genuine information or created information, is from the genuine dispersion [3].

cGANs are flexible and general-purpose arrangements for a wide range of I2I interpretation issues. Not at all like conventional approaches, cGANs are proficient at learning does not fair the mapping from input to yield pictures, but moreover the misfortune work required to prepare this mapping viably. The engineering of cGANs includes a generator arranging mindful for making pictures and a discriminator arrange entrusted with recognizing between genuine and synthesized pictures. Outstandingly, the era of yield pictures in cGANs is conditioned on an input picture, checking a critical flight from unlimited GANs.

A key quality of cGANs, lies in their versatility and adequacy over differing assignments such as photo blend from name maps, question reproduction from edge maps, and picture colorization. The examiners display designing progressions in cGANs, tallying a "U-Net"-based building for the generator and a convolutional “PatchGAN” classifier for the discriminator. These plan choices essentially upgrade the model's capacity to handle different sorts of picture interpretation assignments with higher authenticity. The system utilizes organized misfortune work, combining conventional misfortunes (like L1 or L2) with ill-disposed misfortunes, driving to more practical picture era than minimization of conventional misfortune capacities alone [4].

WGAN has enhanced our understanding of the challenges with traditional GANs, such as poor convergence and mode collapse, and has offered solutions to improve the quality of the generated outputs. Developed to rectify shortcomings linked to the Jensen-Shannon (JS) divergence in GANs, WGAN introduces several benefits. These include the capability to measure distance between distributions that do not intersect, superior smoothing characteristics, and mitigation of the vanishing gradient issue. Additionally, WGAN improves the training stability of GANs and enhances the diversity of the generated samples. This model simplifies the training dynamics between the generator and discriminator, eliminating the need for meticulous tuning [5].

DualGAN presents a bidirectional learning model for image-to-image translation, drawing inspiration from dual learning techniques in natural language processing and adapting them for the image translation field. Unlike the original NLP dual learning method that relies on pre-trained language models, DualGAN utilizes adversarially trained GAN discriminators alongside translators to effectively capture domain distributions. The structure of DualGAN comprises two sets of GANs: the primal GAN and the dual GAN. The primal GAN is tasked with learning a generator GA: U→V and a discriminator DA, while the dual GAN learns a generator GB: V→U and a discriminator DB. These generators and discriminators collaborate to translate images between two domains, establishing a cyclical translation loop. In its operational mechanics, DualGAN's discriminators aim to distinguish between generated (fake) and authentic (real) samples. The model adopts a loss structure influenced by the Wasserstein GAN (WGAN), which enhances the performance in generator

Convergence, sample quality, and optimization stability over the traditional sigmoid cross-entropy loss seen in conventional GANs. A notable attribute of DualGAN is its unsupervised approach, positioning it as a versatile and general-purpose method for image-to-image translation [6].

2.2. Literature Review

Semi-supervised Semantic Segmentation via Image-to-Image Translation tackles the complex issue of semantic segmentation by merging supervised and unsupervised learning techniques. Their innovative two-pass strategy effectively uses unsupervised models to generate pseudo labels and supervised models in conjunction with actual data. This method stands out for its clever integration of both real and pseudo-labeled samples, providing a strong framework that boosts semantic segmentation accuracy. The significance of their work to our research is highlighted by their exploration of semi-supervised learning strategies, particularly in enhancing label accuracy in semantic segmentation. Wu and You's method offers vital insights into how labeled and unlabeled data can be effectively combined, aligning closely with our research on improving DualGAN's ability to translate images with semantic-based labels. Their results are crucial in demonstrating how the integration of pseudo and real labels can enhance the accuracy of semantic information in I2I translation, which is central to this research [2].

The paper [3] investigate unsupervised image translation for semantic segmentation adaptation using the FCN8s-VGG16 architecture. They used a technique called Simplified Unsupervised Image Translation (SUIT) focusing on transforming synthetic images to resemble real images, aimed at improving semantic segmentation without supervised training. They utilized adversarial training to guarantee the generation of realistic images, in addition to implementing a unique semantic-content loss mechanism to uphold semantic consistency between the original and translated images. This method allows for the segmentation model to adapt to real-world data, leveraging the transformed images for enhanced performance in semantic segmentation tasks, without the need for labeled real-world images. Their method achieves significant improvements in multiple semantic segmentation adaptation benchmarks, highlighting the potential of unsupervised methods in this field.

In this paper [4], author explore an innovative approach to image- to-image translation that

emphasizes preserving the intrinsic semantics of images, even amidst geometric variations between the source and target domains. They present a GAN strategy that leverages semantic information encoded as segmentation masks. A key technical advancement is the development of an encoder-decoder generator architecture, which concurrently processes both the image and its semantic details, achieving simultaneous translation to the target domain. To ensure the fidelity of semantic label maps across domains, they incorporate additional losses focused on object configuration and semantic consistency. Demonstrations of this method on various object transformation and domain transfer tasks indicate superior performance in maintaining image semantics over other leading image-to-image translation techniques.

In the exploration of semi-supervised learning techniques, the groundbreaking paper 'Semi-Supervised Learning with Generative Adversarial Networks' offers valuable insights relevant to our research on optimizing unsupervised image-to-image translation. This study presents an innovative extension of Generative Adversarial Networks (GANs) into the semi-supervised learning domain. The authors modified the traditional GAN architecture by adapting the discriminator to perform an additional task of classifying input into one of the N classes, with an extra class for generative outputs. This adaptation not only enhances the data efficiency in classification tasks but also leads to the generation of higher quality samples compared to standard GANs. Crucially, the semi-supervised GAN (SGAN) model underscores the possibility of simultaneous learning of a generative model and a classifier, a concept that aligns seamlessly with the dual objectives of maintaining semantic integrity and optimizing translation quality in unsupervised image-to-image translation. The training algorithm outlined in the paper, which leverages gradient descent on the discriminator/classifier’s outputs, offers a practical framework that could be instrumental in implementing the 'warm start' strategy in our research. This paper's findings and methodologies provide a robust theoretical and practical foundation for employing labeled data in enhancing the performance of unsupervised image-to-image translation, particularly in the context of semantic-based techniques [5]

CHAPTER 3: METHODOLOGY

3.1 Description of Methodology

The purposed methodology is as:

- The process starts by gathering datasets. These datasets include several image-translation domains, such as: Photo → Sketch, Label → Facades & Day → Night
- Collected images undergo standardized preprocessing steps to ensure consistency and improve model performance
- The processed data is divided into paired data & unpaired data.
- The training pipeline involves adjusting critical hyper parameters to optimize performance
- A preliminary training stage uses paired (aligned) examples to teach GAN_A and GAN_B the basic relationship between domains A and Stabilize early learning and initialize generators (G_A, G_B) and discriminators (D_A, D_B)
- After warm-start training, the model transitions to full unsupervised learning using unpaired data in the DualGAN structure.
- Model performance at each hyper parameter setting is monitored, and the combination producing the best results is chosen
- The trained model is evaluated using Mean Squared Error, Fréchet Inception Distance and Structural Similarity Index.
- The optimized model is compared against the baseline original DualGAN to assess improvements in translation quality, reductions in error and better preservation of structure and realism.

3.2 Flowcharts Depicting Methodology

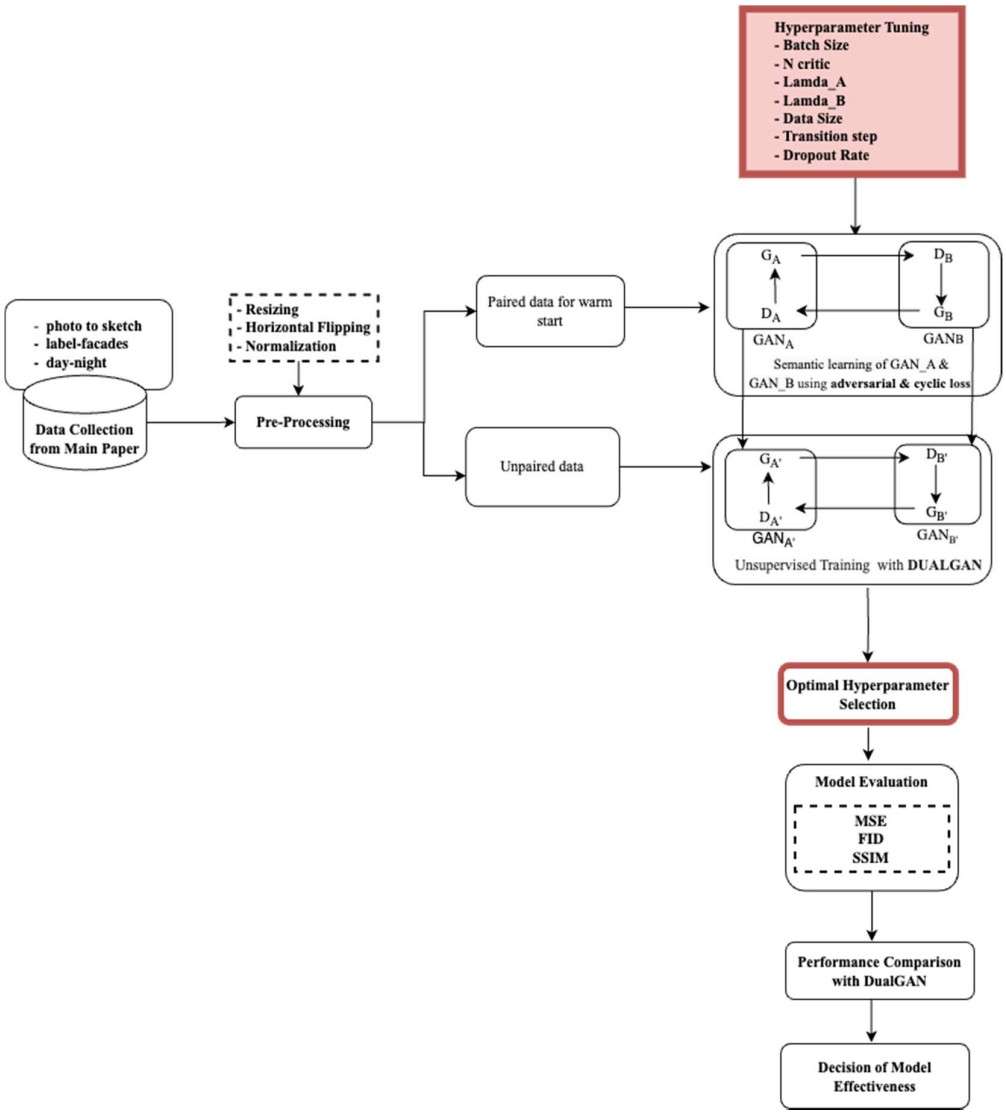


Figure 3. 1. Flowcharts of Depicting Methodology

3.3 Description of Data

The PHOTO–SKETCH dataset consists of 995 training and 199 test grayscale images with 256 × 256 dimensions that are coupled with photographs and sketches [13, 14]. The DAY–NIGHT dataset, which consists of 80 training and 10 test RGB photos of size 512 × 512, provides pairings of scenes taken during the day and at night [9]. The LABEL-FACADES dataset, which consists of 400 training and 100 test RGB images with 256 × 256 dimensions, provides façade images with matching label maps [10]

3.4 Data Pre-processing

The "Modification of DualGAN for Semantic based Image-to-Image Translation through Semi-supervised Learning" paper leverages an innovative approach to data pre-processing to enhance the effectiveness of its model, which is critical when dealing with real-world, unlabeled image datasets.

Data Pre-processing Techniques Used in DualGAN

- **Resizing**: The images are resized to a square format with dimensions of 128 × 128 pixels.
- **Normalization:** It involves adjusting the pixel values of images to fall within the range [-1, 1] by dividing them by 127.5 and then subtracting 1. This standardized procedure aids in stabilizing the learning process of neural networks and accelerates convergence speed.
- **Horizontal Flipping**: It involves randomly flipping images horizontally with a 50% probability during the training phase but not during testing. This data augmentation method enhances the variety of the training dataset, helping the model to better generalize to different image orientations.
- **Supervised and Unsupervised Split in Training Data**: Within the training set, a further subdivision creates a unique environment to leverage both supervised and unsupervised learning paradigms. By allocating a small percentage of the training data to the supervised subset and a larger percentage to the unsupervised subset (e.g., a 1:9 ratio), this study aims to investigate how such a hybrid learning approach influences the overall efficacy and accuracy of the classifier. This approach enables the model to glean insights from a constrained pool of labeled data while leveraging the abundance of unlabeled data, proving especially advantageous in situations where obtaining labeled data is challenging or costly.

3.5 Description of Algorithms/ Mathematical Models

3.5.1. DualGan

The Dual Generative Adversarial Network**,** or DualGAN, stands as a pivotal development in the realm of image-to-image translation, particularly notable for its efficacy in environments where paired training data is scarce or unavailable. This innovative model is engineered to adeptly navigate the challenges of image processing allows it to perform complex translation tasks between two domains, referred to as Domain A and Domain B. This capability makes it a versatile tool for applications ranging from artistic image transformation to practical enhancements.

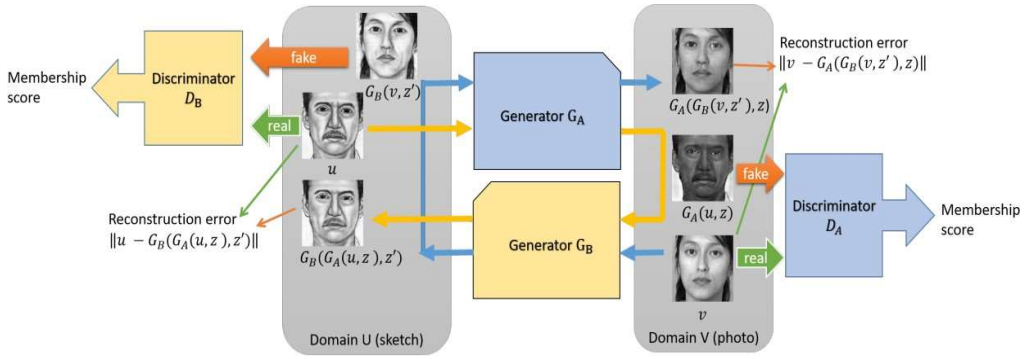


Figure 3. 2.Network Architecture and data flow chart of Dual-GAN for I2I translation

Architecture

The DUALGAN consists of two sets of generators and discriminators, one for each domain (A and B).

Generators

They are responsible for the translation between domains. The A-to-B generator (A2B) translates images from domain A to domain B, and the B-to-A generator (B2A) does the opposite.

Discriminators

Each domain has a discriminator that learns to distinguish between real and generated (fake) images within that domain. The A discriminator (Ad) differentiates between real images from domain A and fake images generated by B2A, while the B discriminator (Bd) differentiates between real images from domain B and fake images generated by A2B.

Significantly, DualGAN integrates both supervised and unsupervised learning approaches. The supervised learning aspect is leveraged during the initial training phase using labeled data, establishing a foundational model. Subsequently, the model transitions to unsupervised learning, handling unpaired and unlabeled data. This combination of learning approaches enhances DualGAN's ability to adapt and perform efficiently in diverse image translation tasks, making it an advanced tool in the field of generative adversarial networks.

Training

The process begins with a dataset that is divided into two parts: one for training the model and the other for testing its performance. The training process of DualGAN model continue using both labeled (paired) and unlabeled (unpaired) data:

Initial Training Phase

- The model starts with supervised learning using paired data (AB).
- Generators A2B and B2A learn to translate images from Domain A to B and B to A using these exact pairs.
- Discriminators Ad (for Domain A) and Bd (for Domain B) learn to differentiate real images from those generated by the generators.

Transition Phase

- After the initial training, there is a decision point or transition step.
- If the transition step is greater than the predefined transition step, we shift the training from supervised learning to unsupervised phase.
- In supervised learning the model learns from the labelled data from warm start phase Which is intended to improve its performance.

Unsupervised Training Phase

- The unsupervised learning phase is now trained with unpaired data
- Generators A2B and B2A adapt to translate images without direct pairings.
- Discriminators Ad and Bd refine their ability to distinguish real from generated images across a broader range of images.

Final DualGAN Model

- The final model integrates learnings from both supervised and unsupervised.
- It becomes capable of translating images between domains A and B, both with and without direct pairings.

Pseudocode of DualGan Algorithm:

Initialize G_XY, G_YX, D_X, D_Y

While training:

x ← sample(X), y ← sample(Y)

y_fake = G_XY(x); x_fake = G_YX(y)

x_cycle = G_YX(y_fake); y_cycle = G_XY(x_fake)

Update D_X, D_Y using real and fake samples

Update G_XY, G_YX using adversarial + cycle losses

End

Mathematically,

Adversarial losses

Each generator tries to fool its corresponding discriminator:

$$\mathcal{L}_{GAN}(G_{AB}, D_B) = \mathbb{E}_b[\log D_B(b)] + \mathbb{E}_a\left[\log\left(1 - D_B\left(G_{AB}(a)\right)\right)\right] \mathcal{L}_{GAN}(G_{BA}, D_A) = \mathbb{E}_a[\log D_A(a)] + \mathbb{E}_b\left[\log\left(1 - D_A\left(G_{BA}(b)\right)\right)\right] \quad (1)$$

Dual (cycle) reconstruction loss

Each image should return to itself after forward and backward translation:

$$= \mathbb{E}_a\left\|G_{BA}\left(G_{AB}(a)\right) - a\right\|_1 + \mathbb{E}_b\left\|G_{AB}\left(G_{BA}(b)\right) - b\right\|_1 \quad (2)$$

Total objective

$$\mathcal{L} = \mathcal{L}_{GAN}(G_{AB}, D_B) + \mathcal{L}_{GAN}(G_{BA}, D_A) + \lambda\mathcal{L}_{dual} \quad (3)$$

3.6. Performance Evaluation Metrics

3.6.1 Mean Square Error (MSE)

It calculates the average squared difference between the original and distorted images. Mathematically, it entails summing the squared differences between the pixels of the two images and then dividing by the total number of pixels [15].

$$\text{MSE} = \frac{1}{mn}\sum_{i=0}^{m-1}\sum_{j=0}^{n-1}[\,I(i,j) - K(i,j)]^2 \tag{4}$$

Where,

- M, n: Dimensions of the images (width and height).
- I (i, j): Pixel value at position (i, j) in the original image.
- K (i,j) : Pixel value at position (i, j) in the GAN-generated image.

The formula calculates the difference between corresponding pixels in the two images, squares this difference (to ensure it's always positive), and then averages these values over the entire image. Since the squared differences are being averaged, the MSE is always non- negative. The range of MSE is theoretically from 0 to positive infinity. A value of 0 indicates a perfect match between the original and reconstructed images, meaning no error. As the MSE value increases, it implies a higher level of dissimilarity or error between the images. In this study, we calculated the mean and median from the values whose mean square error are calculated.

Structural Similarity Index (SSIM):

This metric evaluates the perceived quality of an image by comparing localized patterns of pixel intensities that have been normalized for luminance and contrast. SSIM considers variations in structural information, luminance, and contrast [15].

Luminance, Contrast, and Structure Comparison Functions:

1. Luminance (l(x,y)): Measures the similarity in average brightness between two images.
2. Contrast (c(x,y)): Measures the similarity in the variance of pixel intensities.
3. Structure (s(x,y)): Measures similarity in the structural patterns of the images.

This formula multiplies three comparison functions: luminance, contrast, and structure. It evaluates how similar two images are in terms of their overall appearance, not just pixel-by-pixel differences. In this research, both the Average SSIM (ASSIM) and Median SSIM (MSSIM) were employed to evaluate the similarity between translated images and their corresponding ground truth. ASSIM offered a comprehensive view of the overall image similarity, considering the contributions of all compared patches. MSSIM provided an alternative perspective, being less influenced by potential outliers in individual patch-level comparisons.

Frechet Inception Distance (FID)

The Frechet Inception Distance (FID) stands as a pivotal metric in evaluating the quality of images produced by models like Generative Adversarial Networks (GANs), especially within the realm of image-to-image translation. It assesses the similarity between generated images and real ones by scrutinizing their statistical characteristics. FID achieves this by gauging the distances between the mean and variance of the generated and real images within a deep feature space, typically derived from a layer of the Inception network. The efficacy of the generated images is inversely correlated with the FID score; a lower score indicates that the generated images closely align with the distribution of real images, signaling superior model performance.

The FID score is calculated as follows:

Extract feature vectors from an intermediate layer of the Inception network for both the real and generated images. Calculate the mean and covariance of these feature vectors for both real (μ_r, Σ_r) and generated (μ_g, Σ_g) images [18].

Compute the FID score using the formula:

$$FID = \|\mu_r - \mu_g\| + Tr\,(\sum_r + \sum_g - 2(\sum_r \sum) 2) \quad (5)$$

Where:

- $||\mu r - \mu g||^2$ is the squared Euclidean distance between the mean vectors of real and generated images.
- Tr denotes the trace of the matrix.
- Σr and Σg are the covariance matrices of the real and generated images, respectively. The term 2(Σr Σg /2) involves the square root of the product of the two covariance matrices, which is then multiplied by 2.

CHAPTER 4: WORKING SCHEDULE AND EXPECTED OUTPUT

4.1 Working Schedule

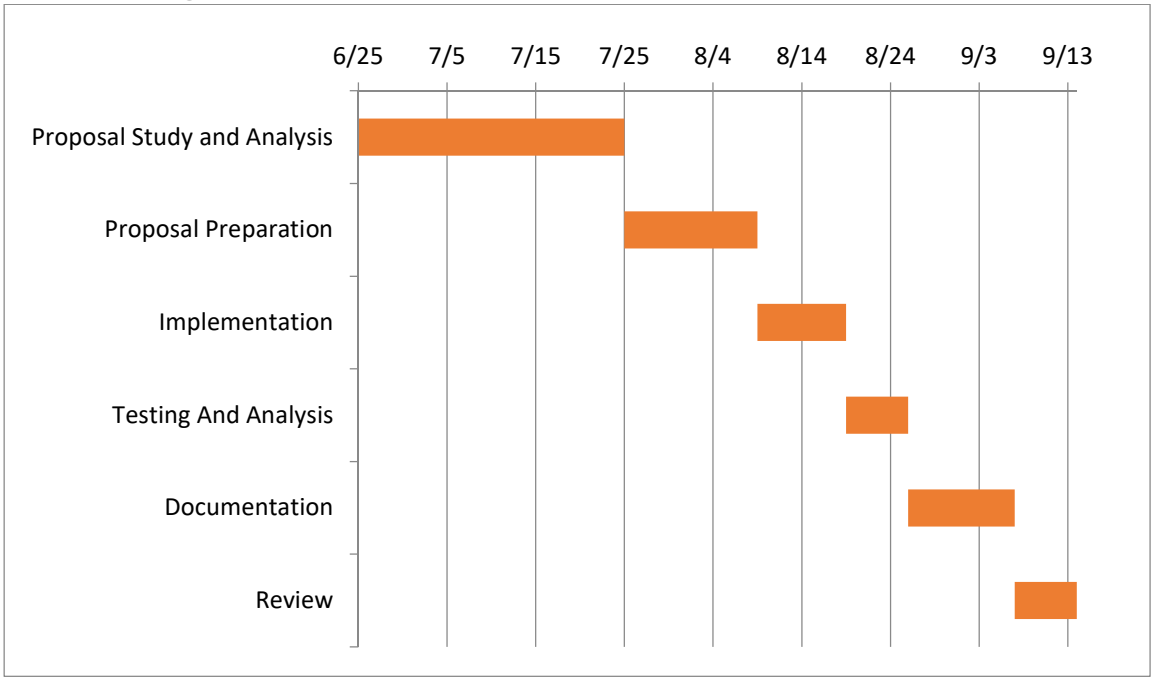


Figure 4. 1.Gantt Chart

4.2 Expected Output

The final model is expected meet the stated objectives with measurable improvements, confirming the potential of labeled data to enhance the capabilities of generative adversarial networks in producing more accurate and visually appealing image translations and to reduce the gap between supervised and unsupervised learning for semantic object segmentation. The model is expected to have support for pixel and label correspondence information in Unsupervised I2I translation. The output of model will be compared with other models described in the literature review section to validate or know the accuracy of the output. The Author wants to test whether that the modification of DualGAN with the integration of labeled data as a warm start can led to performance improvements or not with comparison to DualGAN in image-to-image translation.

References

[1] Isola, Phillip, et al. "Image-to-image translation with conditional adversarial networks." Proceedings of the IEEE conference on computer vision and pattern recognition. 2017.

[2] Z. Yi, H. Zhang, P. Tan, and M. Gong, “DualGAN: Unsupervised Dual Learning for Image-to-Image Translation,” Proceedings of the IEEE International Conference on Computer Vision, Oct. 2017, doi: 10.1109/iccv.2017.310.

[3] A. Creswell, T. White, V. Dumoulin, K. Arulkumaran, B. Sengupta, and A. A. Bharath, “Generative Adversarial Networks: An Overview,” IEEE Signal Processing Magazine, vol. 35, no. 1, pp. 53–65, Jan. 2018, doi: 10.1109/msp.2017.2765202.

[4] P. Isola, J. Zhu, T. Zhou, and A. A. Efros, “Image-to-Image Translation with Conditional Adversarial Networks,” Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, Jul. 2017, doi: 10.1109/cvpr.2017.632.

[5]Y. Cao et al., “Recent advances of generative adversarial networks in computer vision,” IEEE Access, vol. 7, pp. 14985–15006, Jan. 2019, doi: 10.1109/access.2018.2886814.

[6] Yu, Y., Zhang, W. and Deng, Y., 2021. Frechet inception distance (fid) for evaluating gans. China University of Mining Technology Beijing Graduate School: Beijing, China. Vancouver.

[7] Liu, Shuangting, et al. "Pixel level data augmentation for semantic image segmentation using generative adversarial networks." ICASSP 2019-2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, 2019.

[8] Pang, Yingxue, et al. "Image-to-Image Translation: Methods and Applications." arXiv preprint arXiv:2101.08629 (2021).

[9] Cao, Yang-Jie, et al. "Recent advances of generative adversarial networks in computer vision." IEEE Access 7 (2018): 14985-15006.

[10]A. Odena, “Semi-Supervised Learning with Generative Adversarial Networks,” arXiv (Cornell University), Jun. 2016, [Online]. Available: https://arxiv.org/pdf/1606.01583.pdf